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METHODS AND APPARATUS FOR MANUFACTURING A GLASS RIBBON

Abstract

A glass manufacturing apparatus includes a delivery apparatus defining a travel path extending in a travel direction. The delivery apparatus conveys a glass ribbon along the travel path in the travel direction of the delivery apparatus. The glass manufacturing apparatus includes a first forming roll and a second forming roll spaced apart from the first forming roll to define a gap. The first forming roll and the second forming roll receive the glass ribbon along the travel path within the gap. A drive apparatus is coupled to the first forming roll and the second forming roll. The drive apparatus moves one or more of the first forming roll independently of the second forming roll or the second forming roll independently of the first forming roll to change a width of the gap.

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Background/Summary

FIELD [0001] This application is a divisional of U.S. patent application Ser. No. 17/637,661 filed on Sep. 4, 2020, which claims the benefit of priority under 35 U.S.C. § 371 of International Application No. PCT/US2020/049307, filed on Sep. 4, 2020, which claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Application Ser. No. 62/899,450, filed on Sep. 12, 2019, the contents of which are relied upon and incorporated herein by reference in its entirety.

[0002] The present disclosure relates generally to methods for manufacturing a glass ribbon and, more particularly, to methods for manufacturing a glass ribbon with a glass manufacturing apparatus comprising a drive apparatus.
BACKGROUND
[0003] It is known to manufacture molten material into a glass ribbon with a glass manufacturing apparatus. A pair of forming rolls can be spaced apart to define a gap that can receive the molten material. The molten material can pass through the gap, whereupon the molten material can be flattened into a glass ribbon. Gap rings control a width of the gap, and, to adjust the width of the gap, the glass manufacturing apparatus is stopped and the distance separating the forming rolls is adjusted by changing the gap rings. However, temporarily stopping production is inefficient and costly. In addition, variations in the width of the gap may occur, for example, with one end of the forming rolls being closer together than an opposing end.
SUMMARY
[0004] The following presents a simplified summary of the disclosure to provide a basic understanding of some embodiments described in the detailed description.
[0005] There are set forth methods of manufacturing a glass ribbon, comprising introducing a glass ribbon along a travel path in a travel direction to a gap defined between a first forming roll and a second forming roll, and passing the glass ribbon through the gap. Methods comprise changing a width of the gap by moving one or more of the first forming roll independently of the second forming roll. By independently moving the first forming roll, the width of the gap can be changed along a length of the forming rolls, which can adjust for variations in gap width along a length of the forming rolls. In addition, a servo motor can control movement of the first forming roll and/or the second forming roll, in which the servo motor can provide incremental adjustment of the forming rolls and allow for the adjustment to occur without stopping production, thus increasing efficiency. To further reduce variations in the width of the gap, the first forming roll and the second forming roll can be machined after being assembled.
[0006] In some embodiments, a glass manufacturing apparatus comprises a delivery apparatus defining a travel path extending in a travel direction. The delivery apparatus is configured to convey a glass ribbon along the travel path in the travel direction of the delivery apparatus. The glass manufacturing apparatus comprises a first forming roll. The glass manufacturing apparatus comprises a second forming roll spaced apart from the first forming roll to define a gap. The first forming roll and the second forming roll are configured to receive the glass ribbon along the travel path within the gap. The glass manufacturing apparatus comprises a drive apparatus coupled to the first forming roll and the second forming roll. The drive apparatus is configured to move one or more of the first forming roll independently of the second forming roll or the second forming roll independently of the first forming roll to change a width of the gap.
[0007] In some embodiments, the drive apparatus comprises a first translational drive apparatus

coupled to the first forming roll and a second translational drive apparatus coupled to the second forming roll. The first translational drive apparatus is configured to move one or more of a first end or a second end of the first forming roll along a movement axis that is substantially perpendicular to the travel path. The second translational drive apparatus is configured to move one or more of a first end or a second end of the second forming roll along the movement axis.

[0008] In some embodiments, the first forming roll comprises a first outer radial surface that extends about a first forming axis between the first end and the second end of the first forming roll. The first outer radial surface comprises a constant diameter along the first forming axis between the first end and the second end of the first forming roll.

[0009] In some embodiments, the second forming roll comprises a second outer radial surface that extends about a second forming axis between the first end and the second end of the second forming roll. The second outer radial surface comprises a constant diameter along the second forming axis between the first end and the second end of the second forming roll.

[0010] In some embodiments, the glass manufacturing apparatus further comprises a transfer apparatus comprising a frame. The transfer apparatus comprises a first support shaft comprising a first inner end and a first outer end, and a second support shaft comprising a second inner end and a second outer end. The first support shaft and the second support shaft are mounted in the frame. The first forming roll is mounted to the first outer end of the first support shaft and the second outer end of the second support shaft. The transfer apparatus comprises a third support shaft comprising a third inner end and a third outer end, and a fourth support shaft comprising a fourth inner end and a fourth outer end. The third support shaft and the fourth support shaft are mounted in the frame. The second forming roll is mounted to the third outer end of the third support shaft and the fourth outer end of the fourth support shaft.

[0011] In some embodiments, the first inner end and the second inner end are attached to the first translational drive apparatus, and the third inner end and the fourth inner end are attached to the second translational drive apparatus.

[0012] In some embodiments, the transfer apparatus comprises an attachment plate. The third inner end and the fourth inner end are attached to a first side of the attachment plate and the second translational drive apparatus is attached to a second side of the attachment plate. The second translational drive apparatus is configured to move the attachment plate, the third support shaft, and the fourth support shaft along the movement axis.

[0013] In some embodiments, the first support shaft and the second support shaft extend through the attachment plate and move along the movement axis independently of the movement of the attachment plate.

[0014] In some embodiments, the first translational drive apparatus comprises a servo motor.

[0015] In some embodiments, the second translational drive apparatus comprises one or more of a pneumatic cylinder or a servo motor.

[0016] In some embodiments, methods of manufacturing a glass ribbon comprise introducing the glass ribbon along a travel path in a travel direction to a gap defined between a first forming roll and a second forming roll. Methods comprise passing the glass ribbon through the gap. Methods comprise changing a width of the gap by moving one or more of the first forming roll independently of the second forming roll along a movement axis that is substantially perpendicular to the travel path or the second forming roll independently of the first forming roll along the movement axis.

[0017] In some embodiments, methods comprise assembling the first forming roll and the second forming roll, and, after the assembling, machining one or more surfaces of the first forming roll or the second forming roll to reduce a variation in the width of the gap.

[0018] In some embodiments, the changing the width of the gap comprises moving one end of the first forming roll to accommodate for a variation in the width of the gap along the length of the gap.

[0019] In some embodiments, the changing the width of the gap occurs as the glass ribbon is

received within the gap.
[0020] In some embodiments, methods comprise monitoring a characteristic of the glass ribbon and, based on the characteristic, changing the width of the gap.
[0021] In some embodiments, the characteristic comprises one or more of a force exerted on one or more of the first forming roll or the second forming roll, or a thickness of the glass ribbon.
[0022] In some embodiments, methods of manufacturing a glass ribbon comprise assembling a first forming roll and a second forming roll. Methods comprise machining one or more surfaces of the first forming roll or the second forming roll to reduce a variation in a width of a gap defined between the first forming roll and the second forming roll. Methods comprise introducing the glass ribbon along a travel path in a travel direction to the gap. Methods comprise passing the glass ribbon through the gap.
[0023] In some embodiments, the assembling comprises attaching a first shaft to a first side of a first roller and a second shaft to a second side of the first roller to form the first forming roll, and attaching a third shaft to a first side of a second roller and a fourth shaft to a second side of the second roller to form the second forming roll.
[0024] In some embodiments, the assembling comprises attaching the first shaft to a first bearing and the second shaft to a second bearing and attaching the third shaft to a third bearing and the fourth shaft to a fourth bearing.
[0025] In some embodiments, the machining occurs after the assembling of the first forming roll and the second forming roll.
[0026] Additional features and advantages of the embodiments disclosed herein will be set forth in the detailed description that follows, and in part will be clear to those skilled in the art from that description or recognized by practicing the embodiments described herein, including the detailed description which follows, the claims, as well as the appended drawings. It is to be understood that both the foregoing general description and the following detailed description present embodiments intended to provide an overview or framework for understanding the nature and character of the embodiments disclosed herein. The accompanying drawings are included to provide further understanding and are incorporated into and constitute a part of this specification. The drawings illustrate various embodiments of the disclosure, and together with the description explain the principles and operations thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0027] These and other features, embodiments and advantages are better understood when the following detailed description is read with reference to the accompanying drawings, in which:
[0028] FIG. **1** schematically illustrates example embodiments of a glass manufacturing apparatus in accordance with embodiments of the disclosure;
[0029] FIG. **2** illustrates an enlarged view of a drive apparatus taken at view **2** of FIG. **1** in accordance with embodiments of the disclosure;
[0030] FIG. **3** illustrates an enlarged view of an end of a drive apparatus taken at view **3** of FIG. **2** in accordance with embodiments of the disclosure;
[0031] FIG. **4** illustrates a side view of example embodiments of a drive apparatus along line **4-4** of FIG. **3** in accordance with embodiments of the disclosure;
[0032] FIG. **5** illustrates a side view of example embodiments of a drive apparatus along line **5-5** of FIG. **2** in accordance with embodiments of the disclosure;
[0033] FIG. **6** illustrates a top view of example embodiments of a drive apparatus along line **6-6** of FIG. **4** in accordance with embodiments of the disclosure;
[0034] FIG. **7** illustrates a top view of example embodiments of a drive apparatus along line **7-7** of

FIG. **4** in which a first forming roll is movable relative to a second forming roll in accordance with embodiments of the disclosure;

[0035] FIG. **8** illustrates a top view of example embodiments of a drive apparatus similar to FIG. **6** in which a second forming roll is movable relative to a first forming roll in accordance with embodiments of the disclosure;

[0036] FIG. **9** illustrates an exploded view of example embodiments of a forming roll in accordance with embodiments of the disclosure;

[0037] FIG. **10** illustrates a perspective view of example embodiments of a forming roll comprising a roller attached to a first shaft and a second shaft in accordance with embodiments of the disclosure; and

[0038] FIG. **11** illustrates a perspective view of example embodiments of a forming roll comprising a first shaft attached to a first bearing block and a second shaft attached to a second bearing block in accordance with embodiments of the disclosure.

DETAILED DESCRIPTION

[0039] Embodiments will now be described more fully hereinafter with reference to the accompanying drawings in which example embodiments are shown. Whenever possible, the same reference numerals are used throughout the drawings to refer to the same or like parts. However, this disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein.

[0040] The present disclosure relates to a glass manufacturing apparatus and methods for forming a glass ribbon. For purposes of this application, “glass ribbon” may be considered one or more of a glass ribbon in a viscous state, a glass ribbon in an elastic state (e.g., at room temperature) and/or a glass ribbon in a viscoelastic state between the viscous state and the elastic state. Methods and apparatus for forming a glass ribbon will now be described by way of example embodiments. As schematically illustrated in FIG. **1**, in some embodiments, an exemplary glass manufacturing apparatus **100** can comprise a delivery apparatus **101** with a delivery slot **103** to slot draw a stream of glass material, for example, a glass ribbon **105**. In some embodiments, the delivery apparatus **101** can comprise a delivery tube **107** terminating at a lower end **109** in the delivery slot **103**. For example, the delivery tube **107** may comprise a passageway through which the glass-forming material **105** can exit the delivery apparatus **101**. The delivery slot **103** may comprise an opening, a hole etc. through which the glass ribbon **105** can exit the delivery tube **107**. In some embodiments, the delivery tube **107** can be oriented along a direction of gravity, such that the glass ribbon **105** can flow downwardly along the direction of gravity through the delivery tube **107**.

[0041] In some embodiments, the delivery apparatus **101** can define a travel path **111** extending in a travel direction **113** to a forming apparatus **115**. The delivery apparatus **101** can convey the glass ribbon **105** along the travel path **111** in the travel direction **113** of the delivery apparatus **101**. The forming apparatus **115** can comprise a pair of opposing forming rolls, for example, a first forming roll **117** and a second forming roll **119**. In some embodiments, the second forming roll **119** may be spaced apart from the first forming roll **117** to define a gap **121**. In some embodiments, the first forming roll **117** and the second forming roll **119** can rotate counter to one another. For example, in the orientation shown in FIG. **1**, the first forming roll **117** can rotate in a clockwise direction while the second forming roll **119** can rotate in a counterclockwise direction. In some embodiments, the first forming roll **117** and the second forming roll **119** can receive the glass ribbon **105** along the travel path **111** within the gap **121**. The glass ribbon **105** can accumulate between the first forming roll **117** and the second forming roll **119**, whereupon the first forming roll **117** and the second forming roll **119** can flatten, thin, and smooth the glass ribbon **105** into a glass ribbon **123**.

[0042] In some embodiments, the glass ribbon **123** can exit the first forming roll **117** and the second forming roll **119** and may be delivered to a pair of pulling rolls **125**, **127**. The pulling rolls **125**, **127** can pull downwardly on the glass ribbon **123** and, in some embodiments, can generate a tension in the glass ribbon **123** to stabilize and/or stretch the glass ribbon **123**. In some embodiments, the pulling rolls **125**, **127** can rotate counter to one another. For example, in the orientation shown in FIG. **1**, one pulling roll **125** can rotate in a clockwise direction while the other pulling roll **127** can rotate in a counterclockwise direction. In some embodiments, the glass ribbon **123** can move along the travel path **111** in the travel direction **113**. In some embodiments, the glass ribbon **123** can comprise one or more states of material based on the vertical location of the glass ribbon **123**. For example, at one location (e.g., directly below the forming rolls **117**, **119**), the glass ribbon **123** can comprise a viscous material, while at another location (e.g., directly above the pulling rolls **125**, **127**), the glass ribbon **123** can comprise an amorphous solid in a glassy state. In some embodiments, a drive apparatus **129** can be coupled to the first forming roll **117** and the second forming roll **119**. As described relative to FIGS. **2-11**, the drive apparatus **129** can move one or more of the first forming roll **117** or the second forming roll **119** to adjust a dimension of the gap **121**.

[0043] In some embodiments, methods of manufacturing a glass ribbon can comprise introducing the glass ribbon, for example, a viscous glass material, **105** along the travel path **111** in the travel direction **113** to the gap **121** defined between the first forming roll **117** and the second forming roll **119**. For example, the glass ribbon **105** can exit the delivery apparatus **101** and travel along the travel path **111**. The glass ribbon **105** can travel in the travel direction **113**, which, in some embodiments, may be downwardly along a direction of gravity. In some embodiments, methods of manufacturing a glass ribbon can comprise passing the glass ribbon **105** through the gap **121** to form the glass ribbon **123**. For example, as the glass ribbon **105** passes through the gap **121**, the first forming roll **117** and the second forming roll **119** can flatten, thin, and smooth the glass ribbon **105** into a flattened glass ribbon **123**.

[0044] FIG. **2** illustrates the forming apparatus **115** comprising the drive apparatus **129** coupled to the first forming roll **117** and the second forming roll **119** taken at view **2** of FIG. **1**. While the drive apparatus **129** is illustrated as being coupled to first forming roll **117** and the second forming roll **119** in FIG. **1**, the drive apparatus **129** is not so limited. For example, in some embodiments, in addition or in the alternative, the drive apparatus **129** can be coupled to the pulling rolls **125**, **127**. The drive apparatus **129** can control the pulling rolls **125**, **127** in a substantially identical manner as described herein relative to the drive apparatus **129** controlling the first forming roll **117** and the second forming roll **119**. In some embodiments, the first forming roll **117** can comprise a first outer radial surface **201** that extends about a first forming axis **203** between a first end **205** and a second end **207** of the first forming roll **117**. In some embodiments, the first outer radial surface **201** can comprise a constant diameter along the first forming axis **203** between the first end **205** and the second end **207**. In some embodiments, the first forming roll **117** can be attached to one or more bearing blocks, for example, a first bearing block **211** and a second bearing block **213**. The first end **205** of the first forming roll **117** can be attached to the first bearing block **211** and the second end **207** of the first forming roll **117** can be attached to the second bearing block **213**. In some embodiments, the first bearing block **211** and the second bearing block **213** can comprise one or more structures that can facilitate rotation of the first forming roll **117**. For example, the first bearing block **211** and the second bearing block **213** can comprise bearings, for example, spherical bearings, that can allow for rotation of the first forming roll **117** about the first forming axis **203** relative to the first bearing block **211** and the second bearing block **213**. While allowing for rotation of the first forming roll **117**, the first bearing block **211** and the second bearing block **213** can limit inadvertent movement of the first forming roll **117** in or more of the x-direction, y-direction, z-direction, or combinations thereof.

[0045] The second forming roll **119** can be substantially identical to the first forming roll **117**. For example, in some embodiments, the second forming roll **119** can comprise a second outer radial surface **221** that extends about a second forming axis **223** between a first end **225** and a second end **227** of the second forming roll **119**. In some embodiments, the second outer radial surface **221** can comprise a constant diameter along the second forming axis **223** between the first end **225** and the second end **227**. In some embodiments, the first outer radial surface **201** and the second outer radial surface **221** are not limited to comprising a constant diameter. Rather, in some embodiments, one or more of the first outer radial surface **201** or the second outer radial surface **221** may comprise a non-constant diameter, for example, with the diameter varying along an axis along which the first outer radial surface **201** and/or the second outer radial surface **221** extends. The second forming axis **223** can be substantially parallel to the first forming axis **203**. In some embodiments, the second forming roll **119** can be attached to one or more bearing blocks, for example, a third bearing block **231** and a fourth bearing block **233**. The first end **225** of the second forming roll **119** can be attached to the third bearing block **231** and the second end **227** of the second forming roll **119** can be attached to the fourth bearing block **233**. In some embodiments, one or more of the first bearing block **211**, the second bearing block **213**, the third bearing block **231**, or the fourth bearing block **233** may be substantially identical. For example, the third bearing block **231** and the fourth bearing block **233** can comprise one or more structures that can facilitate rotation of the second forming roll **119**. For example, the third bearing block **231** and the fourth bearing block **233** can comprise bearings, for example, spherical bearings, that can allow for rotation of the second forming roll **119** about the second forming axis **223** relative to the third bearing block **231** and the fourth bearing block **233**. While allowing for rotation of the second forming roll **119**, the third bearing block **231** and the fourth bearing block **233** can limit inadvertent movement of the second forming roll **119** in or more of the x-direction, y-direction, z-direction, or combinations thereof. In some embodiments, the first bearing block **211** and the third bearing block **231** may be located on one side of the forming rolls **117**, **119** (e.g., at the first end **205** of the first forming roll **117** and at the first end **225** of the second forming roll **119**), while the second bearing block **213** and the fourth bearing block **233** may be located on an opposing side of the forming rolls **117**, **119** (e.g., at the second end **207** of the first forming roll **117** and at the second end **227** of the second forming roll **119**).

[0046] In some embodiments, the drive apparatus **129** can comprise one or more translational drive apparatuses, for example, a first translational drive apparatus **241** coupled to the first forming roll **117** and a second translational drive apparatus **243** coupled to the second forming roll **119**. In some embodiments, the first translational drive apparatus **241** can be coupled to the first bearing block **211** and the second bearing block **213**, such that the first translational drive apparatus **241** can control movement of the first end **205** and the second end **207** of the first forming roll **117**. While FIG. **2** illustrates the first translational drive apparatus **241** comprising two translational drive apparatuses in some embodiments, the first translational drive apparatus **241** can comprise one or more translational drive apparatuses. For example, as illustrated in FIG. **2**, the first translational drive apparatus **241** can comprise a first end translational drive apparatus **245** and a second end translational drive apparatus **247**. The first end translational drive apparatus **245** can be coupled to the first bearing block **211** and can move the first bearing block **211** along a movement axis, for example, a first movement axis **251**. The second end translational drive apparatus **247** can be coupled to the second bearing block **213** and can move the second bearing block **213** along a movement axis, for example, a second movement axis **253**. In some embodiments, the first movement axis **251** and the second movement axis **253** may be substantially parallel. In some embodiments, the first end translational drive apparatus **245** can move the first bearing block **211** along the first movement axis **251** in a first direction **261**, which is towards the second forming roll **119**, and/or in a second direction **263**, which is away from the second forming roll **119**. In some embodiments, the second end translational drive apparatus **247** can move the second bearing block **213** along the second movement axis **253** in the first direction **261**, which is towards the second forming roll **119**, and/or in the second direction **263**, which is away from the second forming roll **119**.

[0047] In some embodiments, the first translational drive apparatus **241** (e.g., comprising the first end translational drive apparatus **245** and the second end translational drive apparatus **247**) can independently control the movement of the first end **205** and the second end **207** of the first forming roll **117**. For example, in some embodiments, the first end translational drive apparatus **245** can move the first end **205** in the first direction **261** while the second end translational drive apparatus **247** moves the second end **207** in the second direction **263**. In some embodiments, the first end translational drive apparatus **245** can move the first end **205** in the second direction **263** while the second end translational drive apparatus **247** moves the second end **207** in the first direction **261**. In some embodiments, the first end translational drive apparatus **245** can move the first end **205** a first distance in the first direction **261** and the second end translational drive apparatus **247** can move the second end **207** a second distance (e.g., which may be the same as or different than the first distance) in the first direction **261**. In some embodiments, the first end translational drive apparatus **245** can move the first end **205** a first distance in the second direction **263** and the second end translational drive apparatus **247** can move the second end **207** a second distance (e.g., which may be the same as or different than the first distance) in the second direction **263**.

[0048] The first translational drive apparatus **241** is not limited to comprising the first end translational drive apparatus **245** and the second end translational drive apparatus **247**, but, rather, in some embodiments, the first translational drive apparatus **241** can comprise a single translational drive apparatus. For example, when the first translational drive apparatus **241** comprises a single translational drive apparatus, the first translational drive apparatus **241** can be coupled to the first bearing block **211** and the second bearing block **213**, and can move the first bearing block **211** and the second bearing block **213** along the first movement axis **251** and the second movement axis **253**, respectively, in the first direction **261** or the second direction **263**. In contrast to when the first translational drive apparatus **241** comprises two translational drive apparatuses (e.g., as illustrated in FIG. **2**), when the first translational drive apparatus **241** comprises a single first translational drive apparatus, the first end **205** and the second end **207** of the first forming roll **117** are limited to moving together in the same direction (e.g., the first end **205** and the second end **207** moving together in the first direction **261**, or the first end **205** and the second end **207** moving together in the second direction **263**).

[0049] In some embodiments, the first translational drive apparatus **241** can comprise a servo motor. For example, in some embodiments, when the first translational drive apparatus **241** comprises the first end translational drive apparatus **245** and the second end translational drive apparatus **247**, the first end translational drive apparatus **245** can comprise a servo motor, and the second end translational drive apparatus **247** can comprise a servo motor. The servo motors can provide incremental control of the movement of the first bearing block **211** and the second bearing block **213**. For example, the servo motors can allow for movement of the first bearing block **211** and/or the second bearing block **213** a desired distance while the glass manufacturing apparatus **100** is in operation and the glass ribbon **105** is being delivered to the first forming roll **117** and the second forming roll **119**. The first translational drive apparatus **241** comprising one or more servo motors can therefore provide more accurate control of a position of the first forming roll **117** relative to the second forming roll **119**, and a more accurate gap width between the first forming roll **117** and the second forming roll **119**. In addition, the first translational drive apparatus **241** comprising one or more servo motors can facilitate an adjustment of the gap width between the first forming roll **117** and the second forming roll **119** while the glass manufacturing apparatus **100** is in operation, thus reducing downtime and increasing efficiency.

[0050] Referring to the second translational drive apparatus **243**, in some embodiments, the second translational drive apparatus **243** can be coupled to the third bearing block **231** and the fourth bearing block **233**, such that the second translational drive apparatus **243** can control movement of the first end **225** and the second end **227** of the second forming roll **119**. While FIG. **2** illustrates the second translational drive apparatus **243** comprising two translational drive apparatuses **243**, in some embodiments, the second translational drive apparatus **243** can comprise one or more translational drive apparatuses. For example, as illustrated in FIG. **2**, the second translational drive

apparatus **243** can comprise a third end translational drive apparatus **265** and a fourth end translational drive apparatus **267**. The third end translational drive apparatus **265** can be coupled to the third bearing block **231** and can move the third bearing block **231** along a movement axis, for example, the first movement axis **251**. The fourth end translational drive apparatus **267** can be coupled to the fourth bearing block **233** and can move the fourth bearing block **233** along a movement axis, for example, the second movement axis **253**. In some embodiments, the third end translational drive apparatus **265** can move the third bearing block **231** along the first movement axis **251** in the first direction **261** and/or in the second direction **263**. In some embodiments, the fourth end translational drive apparatus **267** can move the fourth bearing block **233** along the second movement axis **253** in the first direction **261** and/or in the second direction **263**.
[0051] In some embodiments, the second translational drive apparatus **243** (e.g., comprising the third end translational drive apparatus **265** and the fourth end translational drive apparatus **267**) can independently control the movement of the first end **225** and the second end **227** of the second forming roll **119**. For example, in some embodiments, the third end translational drive apparatus **265** can move the first end **225** in the first direction **261** while the fourth end translational drive apparatus **267** moves the second end **227** in the second direction **263**. In some embodiments, the third end translational drive apparatus **265** can move the first end **225** in the second direction **263** while the fourth end translational drive apparatus **267** moves the second end **227** in the first direction **261**. In some embodiments, the third end translational drive apparatus **265** can move the first end **225** a first distance in the first direction **261** and the fourth end translational drive apparatus **267** can move the second end **227** a second distance (e.g., which may be the same as or different than the first distance) in the first direction **261**. In some embodiments, the third end translational drive apparatus **265** can move the first end **225** a first distance in the second direction **263** and the fourth end translational drive apparatus **267** can move the second end **227** a second distance (e.g., which may be the same as or different than the first distance) in the second direction **263**.
[0052] The second translational drive apparatus **243** may not be limited to comprising the third end translational drive apparatus **265** and the fourth end translational drive apparatus **267**, but, rather, in some embodiments, the second translational drive apparatus **243** can comprise a single translational drive apparatus. For example, when the second translational drive apparatus **243** comprises a single translational drive apparatus, the second translational drive apparatus **243** can be coupled to the first bearing block **211** and the second bearing block **213**, and can move the first bearing block **211** and the second bearing block **213** along the first movement axis **251** and the second movement axis **253**, respectively, in the first direction **261** or the second direction **263**. In contrast to when the second translational drive apparatus **243** comprises two translational drive apparatuses (e.g., as illustrated in FIG. **2**), when the second translational drive apparatus **243** comprises a single first translational drive apparatus, the first end **225** and the second end **227** of the first forming roll **117** are limited to moving together in the same direction (e.g., the first end **225** and the second end **227** moving together in the first direction **261**, or the first end **225** and the second end **227** moving together in the second direction **263**).
[0053] In some embodiments, the second translational drive apparatus **243** can comprise one or more of a pneumatic cylinder or a servo motor. For example, in some embodiments, when the second translational drive apparatus **243** comprises the third end translational drive apparatus **265** and the fourth end translational drive apparatus **267**, the third end translational drive apparatus **265** can comprise one or more of a pneumatic cylinder or a servo motor, and the fourth end translational drive apparatus **267** can comprise one or more of a pneumatic cylinder or a servo motor. While the pneumatic cylinder may provide less incremental control of the movement of the third bearing block **231** and the fourth bearing block **233**, the first translational drive apparatus **241** can comprise a servo motor which can offset any imprecision in the position of the third bearing block **231** and the fourth bearing block **233**. For example, in some embodiments, the pneumatic cylinder can

adjust the second forming roll **119** between two positions (e.g., a first, or opened, position, and a second, or closed, position). The first, or opened, position may be spaced farther apart from the first forming roll **117** than the second, or closed, position. In operation, the second translational drive apparatus **243** comprising the one or more pneumatic cylinders can maintain the second forming roll **119** in the second, or closed, position. To adjust a width of the gap between the first forming roll **117** and the second forming roll **119**, the servo motors of the first translational drive apparatus **241** can incrementally move the first forming roll **117** relative to the second forming roll **119** while a position of the second forming roll **119** is maintained. In some embodiments, the pneumatic cylinder has at least some degree of flexibility or “give” in response to an increased force applied to the second forming roll **119**. For example, in some embodiments, a solidified piece of material within the glass ribbon **105** may inadvertently be between the first forming roll **117** and the second forming roll **119**, wherein the solidified piece has a size that is larger than a gap width between the first forming roll **117** and the second forming roll **119**. The solidified piece may apply a force upon the first forming roll **117** and the second forming roll **119**. The pneumatic cylinders can allow the second forming roll **119** to move in the first direction **261** and allow for the solidified piece to pass through the gap, thus reducing damage to the first forming roll **117** and the second forming roll **119**.

[0054] In some embodiments, the second translational drive apparatus **243** may not be limited to comprising a pneumatic cylinder, and, instead, can comprise a servo motor. For example, in some embodiments, when the second translational drive apparatus **243** comprises the third end translational drive apparatus **265** and the fourth end translational drive apparatus **267**, the third end translational drive apparatus **265** can comprise a servo motor, and the fourth end translational drive apparatus **267** can comprise a servo motor. The servo motors can provide incremental control of the movement of the third bearing block **231** and the fourth bearing block **233**. For example, the servo motors can allow for movement of the third bearing block **231** and/or the fourth bearing block **233** a desired distance while the glass manufacturing apparatus **100** is in operation and the glass ribbon **105** is being delivered to the first forming roll **117** and the second forming roll **119**. The second translational drive apparatus **243** comprising one or more servo motors can therefore provide more accurate control of a position of the second forming roll **119** relative to the first forming roll **117**, and a more accurate gap width between the first forming roll **117** and the second forming roll **119**. In addition, the second translational drive apparatus **243** comprising one or more servo motors can facilitate an adjustment of the gap width between the first forming roll **117** and the second forming roll **119** while the glass manufacturing apparatus **100** is in operation, thus reducing downtime and increasing efficiency.

[0055] Referring to FIGS. **2-3**, in some embodiments, the drive apparatus **129** can comprise a transfer apparatus **271** that can be coupled to the first forming roll **117** and the second forming roll **119**. For example, in some embodiments, the transfer apparatus **271** can be coupled to the first translational drive apparatus **241** and the second translational drive apparatus **243** on one side, and to the first forming roll **117** and the second forming roll **119** on an opposite side. The transfer apparatus **271** can transfer output motion from the first translational drive apparatus **241** and the second translational drive apparatus **243** to the first forming roll **117** and the second forming roll **119**, respectively, to cause movement of the first forming roll **117** and the second forming roll **119**. The transfer apparatus **271** can comprise one or more support shafts, for example, a first support shaft **273**, a second support shaft **275**, a third support shaft **277**, a fourth support shaft **279**, a fifth support shaft **281**, and a sixth support shaft **283**. The first support shaft **273**, the third support shaft **277**, and the fifth support shaft **281** may be located on a first side of the first forming roll **117** and the second forming roll **119** (e.g., adjacent to the first end **205** and the first end **225**). The second support shaft **275**, the fourth support shaft **279**, and the sixth support shaft **283** may be located on a second side of the of the first forming roll **117** and the second forming roll **119** (e.g., adjacent to the second end **207** and the second end **227**).

[0056] In some embodiments, methods of manufacturing a glass ribbon can comprise monitoring a characteristic of the ribbon of glass ribbon **123** (e.g., illustrated in FIG. **1**) and, based on the characteristic, changing the width of the gap **121**. For example, in some embodiments, the characteristic can comprise one or more of a force exerted on one or more of the first forming roll **117** or the second forming roll **119**, or a thickness of the glass ribbon **123**. In some embodiments, during operation, the glass ribbon **105** can exert a force upon the first forming roll **117** and the second forming roll **119**. This force can be monitored, for example, with a force monitor **291** (e.g., illustrated in FIGS. **2-3**). The force monitor **291** can be coupled, for example, to the third bearing block **231**, though, in some embodiments, the force monitor **291** can be coupled to one or more of the other bearing blocks, for example, the first bearing block **211**, the second bearing block **213** (e.g., illustrated in FIG. **2**), or the fourth bearing block **233** (e.g., illustrated in FIG. **2**). The force monitor **291** can monitor the force that is exerted upon the second forming roll **119** by the glass ribbon **105**. In some embodiments, based on the force that is detected by the force monitor **291**, the width of the gap **121** can be changed. For example, in some embodiments, when the force detected by the force monitor **291** is larger than a predetermined force range, then the width of the gap **121** can be increased by moving the first forming roll **117** and the second forming roll **119** apart. In some embodiments, when the force detected by the force monitor **291** is smaller than a predetermined force range, then the width of the gap **121** can be decreased by moving the first forming roll **117** and the second forming roll **119** closer together.

[0057] In some embodiments, the characteristic of the glass ribbon **123** is not limited to the force exerted on one or more of the first forming roll **117** or the second forming roll **119**. Rather, in some embodiments, the characteristic can comprise a thickness of the glass ribbon **123**. The thickness of the glass ribbon **123** can be monitored, for example, with a visual inspection device, by an operator, etc. During the monitoring, when a thickness of the glass ribbon **123** is larger than a predetermined thickness range, then the width of the gap **121** can be decreased by moving the first forming roll **117** and the second forming roll **119** closer together, which can cause a reduction in the thickness of the glass ribbon **123**. In some embodiments, when the thickness of the glass ribbon **123** is smaller than a predetermined thickness range, then the width of the gap **121** can be increased by moving the first forming roll **117** and the second forming roll **119** apart, which can cause an increase in the thickness of the glass ribbon.

[0058] FIG. **3** illustrates a portion of the transfer apparatus **271** taken at view **3** of FIG. **2**. In some embodiments, the transfer apparatus **271** can comprise a frame **301**. The frame **301** can be positioned between the first forming roll **117** and the drive apparatuses (e.g., the first end translational drive apparatus **245** and the third end translational drive apparatus **265**). In some embodiments, the frame **301** can comprise a wall defining one or more openings, for example, a first opening **303**, a second opening **305**, and a third opening **307**. In some embodiments, the openings of the frame **301** can receive the support shafts. For example, the first opening **303** can receive the first support shaft **273**, the second opening **305** can receive the third support shaft **277**, and the third opening **307** can receive the fifth support shaft **281**. In some embodiments, the first opening **303**, the second opening **305**, and the third opening **307** may be larger in cross-sectional size than the first support shaft **273**, the third support shaft **277**, and the fifth support shaft **281**. For example, the first opening **303** may comprise a cross-sectional size (e.g., diameter in FIG. **3**) that may be larger than a cross-sectional size (e.g., diameter in FIG. **3**) of the first support shaft **273**. The second opening **305** may comprise a cross-sectional size (e.g., diameter in FIG. **3**) that may be larger than a cross-sectional size (e.g., diameter in FIG. **3**) of the third support shaft **277**. The third opening **307** may comprise a cross-sectional size (e.g., diameter in FIG. **3**) that may be larger than a cross-sectional size (e.g., diameter in FIG. **3**) of the fifth support shaft **281**. In some embodiments, due to the first opening **303** being larger than the first support shaft **273**, the first support shaft **273** can be mounted in the frame **301**, for example, by being horizontally and slidingly mounted. In some embodiments, due to the second opening **305** being larger than the third support shaft **277**, the third support shaft **277** can be mounted in the frame **301**, for example, by being horizontally and slidingly mounted. In some embodiments, due to the third opening **307** being larger than the fifth support shaft **281**, the fifth support shaft **281** can be mounted in the frame **301**, for example, by being horizontally and slidingly mounted. By being horizontally and slidingly mounted, the first support shaft **273** can move within the first opening **303** relative to the frame **301** along an axis along which the first support shaft **273** extends, the third support shaft **277** can move within the second opening **305** relative to the frame **301** along an axis along which the third support shaft **277** extends, and/or the fifth support shaft **281** can move within the third opening **307** relative to the frame **301** along an axis along which the fifth support shaft **281** extends.

[0059] In some embodiments, the transfer apparatus **271** can comprise an attachment plate, for example, a first attachment plate **309** (e.g., and a second attachment plate **500** illustrated in FIGS. **5-8**). The first attachment plate **309** can be positioned between the frame **301** and the drive apparatuses (e.g., the first end translational drive apparatus **245** and the third end translational drive apparatus **265**). In some embodiments, the first attachment plate **309** can comprise a wall that extends substantially parallel to the frame **301**. The first attachment plate **309** can be spaced apart from the frame **301**, with the first attachment plate **309** defining one or more openings through which the first support shaft **273**, the third support shaft **277**, and/or the fifth support shaft **281** extend.

[0060] FIG. **4** illustrates a side view of the drive apparatus **129** along line **4-4** of FIG. **3**. In some embodiments, the first attachment plate **309** can define one or more openings, for example, a first attachment opening **401**. In some embodiments, the first attachment plate **309** can receive one or more of the support shafts, for example, the first support shaft **273**. For example, the first support shaft **273** can be received within the first attachment opening **401**. In some embodiments, the first attachment opening **401** may comprise a cross-sectional size (e.g., diameter in FIG. **4**) that may be larger than a cross-sectional size (e.g., diameter in FIG. **4**) of the first support shaft **273**. The first support shaft **273** can be mounted in the first attachment plate **309** (e.g., horizontally and slidingly mounted), with the first support shaft **273** being movable within the first attachment opening **401** relative to the first attachment plate **309** along a first movement axis **402** along which the first support shaft **273** extends.

[0061] In some embodiments, the first support shaft **273** can comprise a first inner end **403** and a first outer end **405**, with the first support shaft **273** extending substantially linearly along the first movement axis **402** between the first inner end **403** and the first outer end **405**. The first forming roll **117** can be mounted, for example, rotationally mounted, to the first outer end **405** of the first support shaft **273**. For example, the first outer end **405** can be attached to the first bearing block **211**, for example, by being received within an opening within the first bearing block **211**. The first outer end **405** can be attached to the first bearing block **211** in several ways. For example, in some embodiments, the first outer end **405** can be threaded into the opening in the first bearing block **211**, with the first outer end **405** comprising a male threading that threadingly engages a female threading in the opening of the first bearing block **211**. In some embodiments, the first outer end **405** can be attached by an adhesive and/or a mechanical fastener (e.g., screws, bolts, etc.) to the first bearing block **211**. By being attached to the first bearing block **211**, the first outer end **405** can cause movement of the first bearing block **211**, for example, when the first support shaft **273** moves along the first movement axis **402**. The first forming roll **117** can be mounted, for example, rotationally mounted, to the first bearing block **211**, with the first forming roll **117** rotatable relative to the first bearing block **211**. As such, the first forming roll **117** can be mounted, for example, rotationally mounted, to the first outer end **405** of the first support shaft **273** via the first bearing block **211**.

[0062] In some embodiments, the first inner end **403** can be attached to the first translational drive apparatus **241**, for example, the first end translational drive apparatus **245**. The first inner end **403** can be attached to the first end translational drive apparatus **245** in several ways, for example, by mechanical fasteners, welding, adhesives, threading engagement, etc. In some embodiments, the first end translational drive apparatus **245** can output a translational drive force along the first movement axis **402**, which can cause movement of the first support shaft **273** along the first movement axis **402** in the first direction **261** and/or the second direction **263**. In some embodiments, the first support shaft **273** can extend through the first attachment plate **309** and move along the first movement axis **402** independently of movement, if any, of the first attachment plate **309**. In some embodiments, as the first end translational drive apparatus **245** causes the first support shaft **273** to move along the first movement axis **402**, the first support shaft **273** can cause corresponding movement of the first bearing block **211** along the first movement axis **402**. This movement of the first bearing block **211** can cause movement of the first end **205** of the first forming roll **117**, such that the first translational drive apparatus **241** can cause the first end **205** of the first forming roll **117** to move along the first movement axis **251**, for example, via the movement of the first support shaft **273** and the first bearing block **211**.

[0063] In some embodiments, the third support shaft **277** can comprise a third inner end **413** and a third outer end **415**, with the third support shaft **277** extending substantially linearly along a third movement axis **417** between the third inner end **413** and the third outer end **415**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the third outer end **415** of the third support shaft **277**. For example, the third outer end **415** can be attached to the third bearing block **231**, for example, by being received within an opening **419** within the third bearing block **231**. The third outer end **415** can be attached to the third bearing block **231** in several ways. For example, in some embodiments, the third outer end **415** can be threaded into the opening **419** in the third bearing block **231**, with the third outer end **415** comprising a male threading that threadingly engages a female threading in the opening **419** of the third bearing block **231**. In some embodiments, the third outer end **415** can be attached by an adhesive and/or a mechanical fastener (e.g., screws, bolts, etc.) to the third bearing block **231**. By being attached to the third bearing block **231**, the third outer end **415** can cause movement of the third bearing block **231**, for example, when the third support shaft **277** moves along the third movement axis **417**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the third bearing block **231**, with the second forming roll **119** rotatable relative to the third bearing block **231**. As such, the second forming roll **119** can be mounted, for example, rotationally mounted, to the third outer end **415** of the third support shaft **277** via the third bearing block **231**.

[0064] In some embodiments, the third support shaft **277** and the first forming roll **117** (e.g., attached to the first bearing block **211**) can move relative to and independent of one another. For example, the third support shaft **277** can move relative to the first forming roll **117** (e.g., attached to the first bearing block **211**), while the first forming roll **117** (e.g., attached to the first bearing block **211**) can move relative to the third support shaft **277**. In some embodiments, the first bearing block **211** can define an opening **421** through which the third support shaft **277** can be received and extend through. In some embodiments, the third support shaft **277** may not be attached to the first bearing block **211**, such that the third support shaft **277** and the first bearing block **211** can move independently of one another. For example, the opening **421** in the first bearing block **211** can be larger in cross-sectional size than a cross-sectional size of the third support shaft **277**. As a result, movement of the third support shaft **277** along the first direction **261** and/or the second direction **263** may not cause movement of the first bearing block **211**. In some embodiments, movement of the first bearing block **211** along the first direction **261** and/or the second direction **263** may not cause movement of the third support shaft **277**. In some embodiments, the third inner end **413** can be attached to the first attachment plate **309**. The third inner end **413** can be attached to the first attachment plate **309** in several ways, for example, by mechanical fasteners, welding, adhesives, threading engagement, etc. for example, the third inner end **413** can be attached to a first side **423** of the first attachment plate **309**.

[0065] In some embodiments, the fifth support shaft **281** can comprise a fifth inner end **433** and a fifth outer end **435**, with the fifth support shaft **281** extending substantially linearly along a fifth movement axis **437** between the fifth inner end **433** and the fifth outer end **435**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the fifth outer end **435** of the fifth support shaft **281**. For example, the fifth outer end **435** can be attached to the third bearing block **231**, for example, by being received within an opening **441** within the third bearing block **231**. The fifth outer end **435** can be attached to the third bearing block **231** in several ways. For example, in some embodiments, the fifth outer end **435** can be threaded into the opening **441** in the third bearing block **231**, with the fifth outer end **435** comprising a male threading that threadingly engages a female threading in the opening **441** of the third bearing block **231**. In some embodiments, the fifth outer end **435** can be attached by an adhesive and/or a mechanical fastener (e.g., screws, bolts, etc.) to the third bearing block **231**. By being attached to the third bearing block **231**, the fifth outer end **435** can cause movement of the third bearing block **231**, for example, when the fifth support shaft **281** moves along the fifth movement axis **437**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the third bearing block **231**, with the second forming roll **119** rotatable relative to the third bearing block **231**. As such, the second forming roll **119** can be mounted, for example, rotationally mounted, to the fifth outer end **435** of the fifth support shaft **281** via the third bearing block **231**.
[0066] In some embodiments, the fifth support shaft **281** and the first forming roll **117** (e.g., attached to the first bearing block **211**) can move relative to and independent of one another. For example, the fifth support shaft **281** can move relative to the first forming roll **117** (e.g., attached to the first bearing block **211**), while the first forming roll **117** (e.g., attached to the first bearing block **211**) can move relative to the fifth support shaft **281**. In some embodiments, the first bearing block **211** can define an opening **443** through which the fifth support shaft **281** can be received and extend through. In some embodiments, the fifth support shaft **281** may not be attached to the first bearing block **211**, such that the fifth support shaft **281** and the first bearing block **211** can move independently of one another. For example, the opening **443** in the first bearing block **211** can be larger in cross-sectional size than a cross-sectional size of the fifth support shaft **281**. As a result, movement of the fifth support shaft **281** along the first direction **261** and/or the second direction **263** may not cause movement of the first bearing block **211**. In some embodiments, movement of the first bearing block **211** along the first direction **261** and/or the second direction **263** may not cause movement of the fifth support shaft **281**. In some embodiments, the fifth inner end **433** can be attached to the first attachment plate **309**. The fifth inner end **433** can be attached to the first attachment plate **309** in several ways, for example, by mechanical fasteners, welding, adhesives, threading engagement, etc. for example, the fifth inner end **433** can be attached to the first side **423** of the first attachment plate **309**.
[0067] In some embodiments, the third support shaft **277** and the fifth support shaft **281** can be positioned on opposing sides of the first support shaft **273**, with the third support shaft **277** and the fifth support shaft **281** extending through the first bearing block **211**. In some embodiments, the third support shaft **277** and the fifth support shaft **281** can extend a greater distance from the frame **301** than the first support shaft **273**, due to the first support shaft **273** being attached to the first bearing block **211**. In some embodiments, the third support shaft **277** and the fifth support shaft **281** can be attached to the third bearing block **231** towards a top and a bottom of the third bearing block **231**. Movement of the first attachment plate **309** in the first direction **261** or the second direction **263** can cause movement of the third support shaft **277** and the fifth support shaft **281** in the first direction **261** or the second direction **263**, respectively, thus causing the first end **225** of the second forming roll **119** to move.
[0068] FIG. **5** illustrates a side view of the drive apparatus **129** along line **5-5** of FIG. **2**. In some embodiments, the transfer apparatus **271** can comprise a second attachment plate **500** that may be substantially identical to the first attachment plate **309** (e.g., illustrated in FIGS. **3-4**). The second

attachment plate **500** can define one or more openings, for example, a second attachment opening **501**. In some embodiments, the second attachment plate **500** can receive one or more of the support shafts, for example, the second support shaft **275**, with the second support shaft **275** received within the second attachment opening **501**. In some embodiments, the second attachment opening **501** may comprise a cross-sectional size (e.g., diameter in FIG. **5**) that may be larger than a cross-sectional size (e.g., diameter in FIG. **5**) of the second support shaft **275**. The second support shaft **275** can be mounted in the second attachment plate **500** (e.g., horizontally and slidingly mounted), with the second support shaft **275** being movable within the second attachment opening **501** relative to the second attachment plate **500** along a second movement axis **502** along which the second support shaft **275** extends.

[0069] In some embodiments, the second support shaft **275** can comprise a second inner end **503** and a second outer end **505**, with the second support shaft **275** extending substantially linearly along the second movement axis **502** between the second inner end **503** and the second outer end **505**. The first forming roll **117** can be mounted, for example, rotationally mounted, to the second outer end **505** of the second support shaft **275**. For example, the second outer end **505** can be attached to the second bearing block **213**, for example, by being received within an opening within the second bearing block **213**. The second outer end **505** can be attached to the second bearing block **213** in several ways. For example, in some embodiments, the second outer end **505** can be threaded into the opening in the second bearing block **213**, with the second outer end **505** comprising a male threading that threadingly engages a female threading in the opening of the second bearing block **213**. In some embodiments, the second outer end **505** can be attached by an adhesive and/or a mechanical fastener (e.g., screws, bolts, etc.) to the second bearing block **213**. By being attached to the second bearing block **213**, the second outer end **505** can cause movement of the second bearing block **213**, for example, when the second support shaft **275** moves along the second movement axis **502**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the second bearing block **213**, with the second forming roll **119** rotatable relative to the second bearing block **213**. As such, the second forming roll **119** can be mounted, for example, rotationally mounted, to the second outer end **505** of the second support shaft **275** via the second bearing block **213**.

[0070] In some embodiments, the second inner end **503** can be attached to the first translational drive apparatus **241**, for example, the second end translational drive apparatus **247**. The second inner end **503** can be attached to the second end translational drive apparatus **247** in several ways, for example, by mechanical fasteners, welding, adhesives, threading engagement, etc. In some embodiments, the second end translational drive apparatus **247** can output a translational drive force along the second movement axis **502**, which can cause movement of the second support shaft **275** along the second movement axis **502** in the first direction **261** and/or the second direction **263**. In some embodiments, the second support shaft **275** can extend through the second attachment plate **500** and move along the second movement axis **502** independently of movement, if any, of the second attachment plate **500**. In some embodiments, as the second end translational drive apparatus **247** causes the second support shaft **275** to move along the second movement axis **502**, the second support shaft **275** can cause corresponding movement of the second bearing block **213** along the second movement axis **502**. This movement of the second bearing block **213** can cause movement of the second end **207** of the first forming roll **117**, such that the first translational drive apparatus **241** can cause the second end **207** of the first forming roll **117** to move along the first movement axis **251**, for example, via the movement of the first support shaft **273** and the first bearing block **211**.

[0071] In some embodiments, the fourth support shaft **279** can comprise a fourth inner end **513** and a fourth outer end **515**, with the fourth support shaft **279** extending substantially linearly along a fourth movement axis **517** between the fourth inner end **513** and the fourth outer end **515**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the fourth outer end

515 of the fourth support shaft **279**. For example, the fourth outer end **515** can be attached to the fourth bearing block **233**, for example, by being received within an opening **519** within the fourth bearing block **233**. The fourth outer end **515** can be attached to the fourth bearing block **233** in several ways. For example, in some embodiments, the fourth outer end **515** can be threaded into the opening **519** in the fourth bearing block **233**, with the fourth outer end **515** comprising a male threading that threadingly engages a female threading in the opening **519** of the fourth bearing block **233**. In some embodiments, the fourth outer end **515** can be attached by an adhesive and/or a mechanical fastener (e.g., screws, bolts, etc.) to the fourth bearing block **233**. By being attached to the fourth bearing block **233**, the fourth outer end **515** can cause movement of the fourth bearing block **233**, for example, when the fourth support shaft **279** moves along the fourth movement axis **517**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the fourth bearing block **233**, with the second forming roll **119** rotatable relative to the fourth bearing block **233**. As such, the second forming roll **119** can be mounted, for example, rotationally mounted, to the fourth outer end **515** of the fourth support shaft **279** via the fourth bearing block **233**.

[0072] In some embodiments, the fourth support shaft **279** and the first forming roll **117** (e.g., attached to the second bearing block **213**) can move relative to and independent of one another. For example, the fourth support shaft **279** can move relative to the first forming roll **117** (e.g., attached to the second bearing block **213**), while the first forming roll **117** (e.g., attached to the second bearing block **213**) can move relative to the fourth support shaft **279**. In some embodiments, the second bearing block **213** can define an opening **521** through which the fourth support shaft **279** can be received and extend through. In some embodiments, the fourth support shaft **279** may not be attached to the second bearing block **213**, such that the fourth support shaft **279** and the second bearing block **213** can move independently of one another. For example, the opening **521** in the second bearing block **213** can be larger in cross-sectional size than a cross-sectional size of the fourth support shaft **279**. As a result, movement of the fourth support shaft **279** along the first direction **261** and/or the second direction **263** may not cause movement of the second bearing block **213**. In some embodiments, movement of the second bearing block **213** along the first direction **261** and/or the second direction **263** may not cause movement of the fourth support shaft **279**. In some embodiments, the fourth inner end **513** can be attached to the second attachment plate **500**. The fourth inner end **513** can be attached to the second attachment plate **500** in several ways, for example, by mechanical fasteners, welding, adhesives, threading engagement, etc. for example, the fourth inner end **513** can be attached to a first side **523** of the second attachment plate **500**.

[0073] In some embodiments, the sixth support shaft **283** can comprise a sixth inner end **533** and a sixth outer end **535**, with the sixth support shaft **283** extending substantially linearly along a sixth movement axis **537** between the sixth inner end **533** and the sixth outer end **535**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the sixth outer end **535** of the sixth support shaft **283**. For example, the sixth outer end **535** can be attached to the fourth bearing block **233**, for example, by being received within an opening **541** within the fourth bearing block **233**. The sixth outer end **535** can be attached to the fourth bearing block **233** in several ways. For example, in some embodiments, the sixth outer end **535** can be threaded into the opening **541** in the fourth bearing block **233**, with the sixth outer end **535** comprising a male threading that threadingly engages a female threading in the opening **541** of the fourth bearing block **233**. In some embodiments, the sixth outer end **535** can be attached by an adhesive and/or a mechanical fastener (e.g., screws, bolts, etc.) to the fourth bearing block **233**. By being attached to the fourth bearing block **233**, the sixth outer end **535** can cause movement of the fourth bearing block **23**, for example, when the sixth support shaft **283** moves along the sixth movement axis **537**. The second forming roll **119** can be mounted, for example, rotationally mounted, to the fourth bearing block **233**, with the second forming roll **119** rotatable relative to the fourth bearing block **233**. As such, the second forming roll **119** can be mounted, for example, rotationally mounted, to the sixth outer end **535** of the sixth support shaft **283** via the fourth bearing block **233**.

[0074] In some embodiments, the sixth support shaft **283** and the first forming roll **117** (e.g., attached to the second bearing block **213**) can move relative to and independent of one another. For example, the sixth support shaft **283** can move relative to the first forming roll **117** (e.g., attached to the second bearing block **213**), while the first forming roll **117** (e.g., attached to the second bearing block **213**) can move relative to the sixth support shaft **283**. In some embodiments, the second bearing block **213** can define an opening **543** through which the sixth support shaft **283** can be received and extend through. In some embodiments, the sixth support shaft **283** may not be attached to the second bearing block **213**, such that the sixth support shaft **283** and the second bearing block **213** can move independently of one another. For example, the opening **543** in the second bearing block **213** can be larger in cross-sectional size than a cross-sectional size of the sixth support shaft **283**. As a result, movement of the sixth support shaft **283** along the first direction **261** and/or the second direction **263** may not cause movement of the second bearing block **213**. In some embodiments, movement of the second bearing block **213** along the first direction **261** and/or the second direction **263** may not cause movement of the sixth support shaft **283**. In some embodiments, the sixth inner end **533** can be attached to the second attachment plate **500**. The sixth inner end **533** can be attached to the second attachment plate **500** in several ways, for example, by mechanical fasteners, welding, adhesives, threading engagement, etc. In some embodiments, the sixth inner end **533** can be attached to the first side **523** of the second attachment plate **500**.

[0075] In some embodiments, the fourth support shaft **279** and the sixth support shaft **283** can be positioned on opposing sides of the second support shaft **275**, with the fourth support shaft **279** and the sixth support shaft **283** extending through the second bearing block **213**. In some embodiments, the fourth support shaft **279** and the sixth support shaft **283** can extend a greater distance from the frame **301** than the second support shaft **275**, due to the second support shaft **275** being attached to the second bearing block **213**. In some embodiments, the fourth support shaft **279** and the sixth support shaft **283** can be attached to the fourth bearing block **233** towards a top and a bottom of the fourth bearing block **233**. Movement of the second attachment plate **500** in the first direction **261** or the second direction **263** can cause movement of the fourth support shaft **279** and the sixth support shaft **283** in the first direction **261** or the second direction **263**, respectively, thus causing the second end **227** of the second forming roll **119** to move.

[0076] FIG. **6** illustrates a top-down view of the drive apparatus **129** along line **6-6** of FIG. **4**. In some embodiments, the drive apparatus **129** can independently control the movement of the first end **205** of the first forming roll **117**, the second end **207** of the first forming roll **117**, the first end **225** of the second forming roll **119**, or the second end **227** of the second forming roll **119** in the first direction **261** and/or the second direction **263**. For example, the first attachment plate **309** and the second attachment plate **500** can be spaced apart to define a gap between the first attachment plate **309** and the second attachment plate **500**. The first attachment plate **309** and the second attachment plate **500** can move independently of one another, with the first attachment plate **309** movable in the first direction **261** and/or the second direction **263**, and the second attachment plate **500** movable, independently of the first attachment plate **309**, in the first direction **261** and/or the second direction **263**. In some embodiments, the first attachment plate **309** can comprise the first side **423** and an opposing second side **601**. The third support shaft **277** and the fifth support shaft **281** can be attached to the first side **423** of the first attachment plate **309** (e.g., illustrated in FIG. **4** with the third inner end **413** of the third support shaft **277** and the fifth inner end **433** of the fifth support shaft **281** attached to the first side **423** of the first attachment plate **309**). The second translational drive apparatus **243**, for example, the third end translational drive apparatus **265**, can be attached to the second side **601** of the first attachment plate **309**. In some embodiments, the third end translational drive apparatus **265** can comprise a first drive shaft **603** that may be attached to the second side **601** of the first attachment plate **309**. The third end translational drive apparatus **265** can output movement to cause the first drive shaft **603** to move (e.g., in the first direction **261** and/or the second direction **263**), which can cause movement of the first attachment plate **309**. Due to the attachment of the third support shaft **277** and the fifth support shaft **281** to the first side **423** of the first attachment plate **309**, movement of the first attachment plate **309** can cause movement of the third support shaft **277** and the fifth support shaft **281**, which can cause corresponding movement of the first end **225** of the second forming roll **119**.

[0077] In some embodiments, the second attachment plate **500** can comprise the first side **523** and an opposing second side **605**. The fourth support shaft **279** and the sixth support shaft **283** can be attached to the first side **523** of the second attachment plate **500** (e.g., illustrated in FIG. **5** with the fourth inner end **513** of the fourth support shaft **279** and the sixth inner end **533** of the sixth support shaft **283** attached to the first side **523** of the second attachment plate **500**). The second translational drive apparatus **243**, for example, the fourth end translational drive apparatus **267**, can be attached to the second side **605** of the second attachment plate **500**. In some embodiments, the fourth end translational drive apparatus **267** can comprise a second drive shaft **607** that may be attached to the second side **605** of the second attachment plate **500**. The fourth end translational drive apparatus **267** can output movement to cause the second drive shaft **607** to move (e.g., in the first direction **261** and/or the second direction **263**), which can cause movement of the second attachment plate **500**. Due to the attachment of the fourth support shaft **279** and the sixth support shaft **283** to the first side **523** of the second attachment plate **500**, movement of the second attachment plate **500** can cause movement of the fourth support shaft **279** and the sixth support shaft **283**, which can cause corresponding movement of the second end **227** of the second forming roll **119**.

[0078] FIG. **7** illustrates a top-down view of the drive apparatus **129** along line **7-7** of FIG. **4**. In some embodiments, methods of manufacturing a glass ribbon can comprise changing a width **701** of the gap **121** by moving one or more of the first forming roll **117** independently of the second forming roll **119** along a movement axis **703** that may be substantially perpendicular to the travel path **111** or the second forming roll **119** independently of the first forming roll **117** along the movement axis **703** to change the width **701** of the gap **121**. In some embodiments, the drive apparatus **129** can move one or more of the first forming roll **117** independently of the second forming roll **119** or the second forming roll **119** independently of the first forming roll **117** to change the width **701** of the gap **121**. For example, the first translational drive apparatus **241** can move the first end **205** and/or the second end **207** of the first forming roll **117** independently of the second translational drive apparatus **243** moving the first end **225** and/or the second end **227** of the second forming roll **119**. The first support shaft **273** can be attached to the first bearing block **211** and the first end translational drive apparatus **245**, for example, with the first outer end **405** of the first support shaft **273** attached to the first bearing block **211** and the first inner end **403** of the first support shaft **273** attached to the first end translational drive apparatus **245**. In some embodiments, the first support shaft **273** can extend through openings in the frame **301** and the first attachment plate **309** (e.g., also illustrated in FIG. **4**), such that the first support shaft **273** may move independently of the frame **301** and the first attachment plate **309**. In some embodiments, the first end translational drive apparatus **245** can output movement to cause the first support shaft **273** to move (e.g., in the first direction **261** and/or the second direction **263**). Movement of the first support shaft **273** can cause movement of the first bearing block **211** (e.g., in the first direction **261** and/or the second direction **263**), which can increase or decrease the width **701** of the gap **121** at the first end **205** of the first forming roll **117**.

[0079] In some embodiments, for example, when the first translational drive apparatus **241** comprises a plurality of drive apparatuses (e.g., the first end translational drive apparatus **245** and the second end translational drive apparatus **247**), the first end **205** of the first forming roll **117** and the second end **207** of the first forming roll **117** can move independently of one another. For example, the second support shaft **275** can be attached to the second bearing block **213** and the second end translational drive apparatus **247**, for example, with the second outer end **505** of the second support shaft **275** attached to the second bearing block **213** and the second inner end **503** of the second support shaft **275** attached to the first end translational drive apparatus **245**. In some embodiments, the second support shaft **275** can extend through openings in the frame **301** and the second attachment plate **500** (e.g., also illustrated in FIG. **5**), such that the second support shaft **275** may move independently of the frame **301** and the second attachment plate **500**. In some embodiments, the second end translational drive apparatus **247** can output movement to cause the second support shaft **275** to move (e.g., in the first direction **261** and/or the second direction **263**). Movement of the second support shaft **275** can cause movement of the second bearing block **213** (e.g., in the first direction **261** and/or the second direction **263**), which can increase or decrease the width **701** of the gap **121** at the second end **207** of the first forming roll **117**.

[0080] In some embodiments, the first end translational drive apparatus **245** and the second end translational drive apparatus **247** can operate independently of one another, such that the movement of the first end **205** and the second end **207** of the first forming roll **117** may be independent of one another. For example, in some embodiments, to decrease the width **701** of the gap **121**, the first end translational drive apparatus **245** can move the first support shaft **273** in the first direction **261** while the second end translational drive apparatus **247** can move the second support shaft **275** in the first direction **261**, which can move the first forming roll **117** towards from the second forming roll **119**. In some embodiments, to increase the width **701** of the gap **121**, the first end translational drive apparatus **245** can move the first support shaft **273** in the second direction **263** while the second end translational drive apparatus **247** can move the second support shaft **275** in the second direction **263**, which can move the first forming roll **117** away from the second forming roll **119**. In some embodiments, a diameter of the first forming roll **117** may not be constant along a length of the first forming roll **117**, such that one end of the first forming roll **117** may be spaced a different distance from the second forming roll **119** than the opposing end of the first forming roll **117**. To accommodate for differences in the diameter of the first forming roll **117**, in some embodiments, one end (e.g., the first end **205** or the second end **207**) of the first forming roll **117** can be moved in the first direction **261** and/or the second direction **263** while the opposing end of the first forming roll **117** may remain stationary. As such, the first translational drive apparatus **241** can move one or more of the first end **205** or the second end **207** of the first forming roll **117** along the movement axis **703** that may be substantially perpendicular to the travel path **111**.

[0081] FIG. **8** illustrates a top-down view of the drive apparatus **129** similar to FIG. **6** along line **6-6** of FIG. **4**. In some embodiments, changing the width **701** of the gap **121** is not limited to moving the first forming roll **117** independently of the second forming roll **119** along the movement axis **703** that is substantially perpendicular to the travel path **111**. Rather, in some embodiments, methods of manufacturing a glass ribbon can comprise moving the second forming roll **119** independently of the first forming roll **117** along the movement axis **703**. For example, the second translational drive apparatus **243** can move the first end **225** and/or the second end **227** of the second forming roll **119** independently of the first translational drive apparatus **241** that moves the first end **205** and/or the second end **207** of the first forming roll **117**. The third support shaft **277** and the fifth support shaft **281** can be attached to the third bearing block **231** and the third end translational drive apparatus **265**. In some embodiments, the third support shaft **277** and the fifth support shaft **281** can extend through openings in the frame **301** and can be attached to the first side **423** of the first attachment plate **309** (e.g., also illustrated in FIG. **4**), such that the third support shaft **277** and the fifth support shaft **281** can move independently of the frame **301**. In some embodiments, the third end translational drive apparatus **265** can output movement to cause the third support shaft **277** and the fifth support shaft **281** to move (e.g., in the first direction **261** and/or the second direction **263**). Movement of the third support shaft **277** and the fifth support shaft **281** can cause movement of the third bearing block **231** (e.g., in the first direction **261** and/or the second direction **263**), which can increase or decrease the width **701** of the gap **121** at the first end **225** of the second forming roll **119**.

[0082] In some embodiments, for example, when the second translational drive apparatus **243** comprises a plurality of drive apparatuses (e.g., the third end translational drive apparatus **265** and the fourth end translational drive apparatus **267**), the first end **225** of the second forming roll **119** and the second end **227** of the second forming roll **119** can move independently of one another. For example, the fourth support shaft **279** and the sixth support shaft **283** can be attached to the fourth bearing block **233** and the fourth end translational drive apparatus **267**. In some embodiments, the fourth support shaft **279** and the sixth support shaft **283** can extend through openings in the frame **301** (e.g., also illustrated in FIG. **5**), such that the fourth support shaft **279** and the sixth support shaft **283** may move independently of the frame **301**. In some embodiments, the fourth end translational drive apparatus **267** can output movement to cause the fourth support shaft **279** and the sixth support shaft **283** to move (e.g., in the first direction **261** and/or the second direction **263**). Movement of the fourth support shaft **279** and the sixth support shaft **283** can cause movement of the fourth bearing block **233** (e.g., in the first direction **261** and/or the second direction **263**), which can increase or decrease the width **701** of the gap **121** at the second end **227** of the second forming roll **119**.
[0083] In some embodiments, the third end translational drive apparatus **265** and the fourth end translational drive apparatus **267** can operate independently of one another, such that the movement of the first end **225** and the second end **227** of the second forming roll **119** may be independent of one another. For example, in some embodiments, to decrease the width **701** of the gap **121**, the third end translational drive apparatus **265** can move the third support shaft **277** and the fifth support shaft **281** in the second direction **263** while the fourth end translational drive apparatus **267** can move the fourth support shaft **279** and the sixth support shaft **283** in the second direction **263**, which can move the second forming roll **119** towards the first forming roll **117**. In some embodiments, to increase the width **701** of the gap **121**, the third end translational drive apparatus **265** can move the third support shaft **277** and the fifth support shaft **281** in the first direction **261** while the fourth end translational drive apparatus **267** can move the fourth support shaft **279** and the sixth support shaft **283** in the first direction **261**, which can move the second forming roll **119** away from the first forming roll **117**. In some embodiments, a diameter of the second forming roll **119** may not be constant along a length of the second forming roll **119**, such that one end of the second forming roll **119** may be spaced a different distance from the first forming roll **117** than the opposing end of the second forming roll **119**. To accommodate for differences in the diameter of the second forming roll **119**, in some embodiments, one end (e.g., the first end **225** or the second end **227**) of the second forming roll **119** can be moved in the first direction **261** and/or the second direction **263** while the opposing end of the second forming roll **119** may remain stationary. As such, the second translational drive apparatus **243** can move one or more of the first end **225** or the second end **227** of the second forming roll **119** along the movement axis **703**.
[0084] In some embodiments, methods of manufacturing the glass ribbon can comprise changing the width **701** of the gap **121** by moving one or more of the first forming roll **117** independently of the second forming roll **119** along the movement axis that is substantially perpendicular to the travel path or the second forming roll independently of the first forming roll along the movement axis **703** that is substantially perpendicular to the travel path **111** or the second forming roll **119** independently of the first forming roll **117** along the movement axis **703**. For example, the first translational drive apparatus **241** can control the movement of the first forming roll **117** while the second translational drive apparatus **243** can control movement of the second forming roll **119**. In some embodiments, the first translational drive apparatus **241** can move the first forming roll **117** independently of the second forming roll **119**. In some embodiments, the second translational drive apparatus **243** can move the second forming roll **119** independently of the first forming roll **117**.
[0085] In some embodiments, changing the width **701** of the gap **121** can comprise moving one end of the first forming roll **117** to accommodate for a variation in the width **701** of the gap **121** along the length of the gap **121**. For example, the first translational drive apparatus **241** can comprise the

first end translational drive apparatus **245** and the second end translational drive apparatus **247**, wherein the first end translational drive apparatus **245** can control movement of the first end **205** of the first forming roll **117**, while the second end translational drive apparatus **247** can control movement of the second end **207** of the first forming roll **117**. In some embodiments, the first end translational drive apparatus **245** and the second end translational drive apparatus **247** can work independently of one another, such that the first end **205** and the second end **207** can be adjusted independently of one another. As such, one end of the first forming roll **117** can be moved to accommodate for variations in the width **701** of the gap **121** along the length of the gap **121**. For example, due to the geometry of the first forming roll **117** and/or the second forming roll **119**, the width **701** of the gap **121** at the first end **205** may be greater than the width **701** of the gap **121** at the second end **207**. To accommodate for this variation in the width **701**, the first end translational drive apparatus **245** can move the first end **205** closer to the second forming roll **119** (e.g., in the first direction **261**), thus reducing the variation in the width **701**. In addition, or in the alternative, the second end translational drive apparatus **247** can move the second end **207** away from the second forming roll **119** (e.g., in the second direction **263**), thus reducing the variation in the width **701**. In some embodiments, changing the width **701** of the gap **121** can occur as the glass ribbon **105** (e.g., illustrated in FIG. **1**) is received within the gap **121**. For example, the first translational drive apparatus **241** can comprise one or more servo motors, which can facilitate movement and positional adjustment of the first forming roll **117** during operation and without stopping production.

[0086] FIG. **9** illustrates an exploded view of a forming roll **901**, for example, the first forming roll **117** or the second forming roll **119**. In some embodiments, the forming roll **901** may be substantially identical to the first forming roll **117** and/or the second forming roll (e.g., illustrated in FIGS. **1-8**). In some embodiments, the forming roll **901** can comprise a roller **903**, for example, an insulating cylinder or coating. In some embodiments, the roller **903** can comprise one or more of a stainless-steel material, an Inconel material, or a ceramic coated stainless-steel material. In some embodiments, the roller **903** can comprise a ceramic coating, a sleeve, or other ceramic base materials, for example, zirconia. In some embodiments, the roller **903** can be substantially hollow and may comprise a substantially circular cross-section, for example, a diameter. The roller **903** can extend along a forming axis **905** and may comprise an outer radial surface **907**. In some embodiments, the outer radial surface **907** can comprise a constant diameter along the forming axis **905** between a first end **909** and a second end **911**.

[0087] In some embodiments, the forming roll **901** can comprise one or more shafts, for example, a first shaft **915** and a second shaft **917**. The first shaft **915** can be attached to a first side **919** of the roller **903** while the second shaft **917** can be attached to a second side **921** of the roller **903**. In some embodiments, methods of manufacturing a glass ribbon can comprise assembling the forming roll **901** (e.g., assembling the first forming roll **117** and the second forming roll **119**). For example, assembling the forming roll **901** can comprise attaching the first shaft **915** to the first side **919** of the roller **903** and the second shaft **917** to the second side **921** of the roller **903** to form the forming roll **901**. In some embodiments, the first shaft **915** can comprise a first end cap **923** that can engage with (e.g., contact, be received within, etc.) the first end **909** of the roller **903**. The second shaft **917** can comprise a second end cap **925** that can engage with (e.g., contact, be received within, etc.) the second end **911** of the roller **903**. In some embodiments, the forming roll **901** can comprise one or more fasteners (e.g., screws, bolts, adhesives, etc.) that can attach the first shaft **915** and the second shaft **917** to the roller **903**. The fasteners can maintain the first shaft **915** and the second shaft **917** in attachment to the roller **903** and limit inadvertent detachment of the first shaft **915** or the second shaft **917** from the roller **903**.

[0088] FIG. **10** illustrates the forming roll **901** following the assembly of the forming roll **901** by attaching the first shaft **915** and the second shaft **917** to the roller **903**. After the assembling of the forming roll **901** (e.g., the first forming roll **117** and the second forming roll **119**), methods of manufacturing a glass ribbon can comprise machining one or more surfaces of the forming roll **901** (e.g., the first forming roll **117** and the second forming roll **119**) to reduce a variation in the width (e.g., width **701** illustrated in FIGS. **7-8**) of the gap (e.g., gap **121** illustrated in FIGS. **6-8**) defined between the first forming roll **117** and the second forming roll **119**. For example, it is beneficial for the width **701** of the gap **121** to be substantially constant along a length of the forming roll **901** (e.g., the first forming roll **117** and the second forming roll **119** illustrated in FIGS. **6-8**). To maintain a substantially constant width **701**, accurate dimensions of the roller **903**, the first shaft **915**, and the second shaft **917** may reduce variations in the width **701** of the gap **121**. However, even if relatively accurate dimensions of the dimensions of the roller **903**, the first shaft **915**, and the second shaft **917** are obtained, there may still be variations in dimensions caused by the assembly of the forming roll **901**. To reduce these variations, the forming roll **901** can be machined (e.g., with the machining **1001** illustrated schematically with arrows). For example, the machining **1001** can occur after the assembling of the forming roll **901** (e.g., the first forming roll **117** and the second forming roll **119**), wherein the machining **1001** can comprise grinding, cutting, etc. the one or more surfaces of the forming roll **901**. In some embodiments, the one or more surfaces of the forming roll **901** can comprise, for example, a surface of the first shaft **915**, a surface of the second shaft **917**, and/or an outer radial surface **907** of the roller **903**.

[0089] Referring to FIG. **11**, in some embodiments, the assembling the forming roll **901** can comprise attaching the first shaft **915** to a first bearing **1101** of a first bearing block **1103** and the second shaft **917** to a second bearing **1105** of a second bearing block **1107**. In some embodiments, the first bearing block **1103** and the second bearing block **1107** may be substantially identical to one or more of the first bearing block **211**, the second bearing block **213**, the third bearing block **231**, or the fourth bearing block **233** (e.g., illustrated in FIG. **2**). The first bearing **1101** can be received within the first bearing block **1103**, with the first bearing **1101** defining an opening through which the first shaft **915** can be received. In some embodiments, the first bearing **1101** can facilitate rotation of the first shaft **915** relative to the first bearing block **1103**. The second bearing **1105** can be received within the second bearing block **1107**, with the second bearing **1105** defining an opening through which the second shaft **917** can be received. In some embodiments, the second bearing **1105** can facilitate rotation of the second shaft **917** relative to the second bearing block **1107**. In some embodiments, the machining (e.g., illustrated in FIG. **10**) of the forming roll **901** can occur prior to or after the attaching of the first shaft **915** to the first bearing block **1103** and the second shaft **917** to the second bearing block **1107**. In some embodiments, to accommodate for thermal expansion during operation, one or more of the first bearing **1101** or the second bearing **1105** can be movable relative to the first bearing block **1103** or the second bearing block **1107**, respectively. For example, the first bearing **1101** can be movable relative to the first bearing block **1103** in the first direction **1111** and/or the second direction **1113**, wherein the first direction **1111** and the second direction **1113** may be substantially parallel to the forming axis **905**. In addition, or in the alternative, in some embodiments, the second bearing **1105** can be movable relative to the second bearing block **1107** in the first direction **1111** and/or the second direction **1113**. In some embodiments, the first bearing **1101** and the second bearing **1105** are not limited to facilitating movement along the forming axis **905** (e.g., in the first direction **1111** and/or the second direction **1113**) and about the forming axis (e.g., when the first shaft **915** and the second shaft **917** rotate). Rather, in some embodiments, the first bearing **1101** and/or the second bearing **1105** may be pivotable relative to the first bearing block **1103** and the second bearing block **1107**, respectively. For example, in some embodiments, the first bearing **1101** may be pivotable relative to the first bearing block **1103** and/or the second bearing **1105** may be pivotable relative to the second bearing block **1107**. The pivoting of the first bearing **1101** and/or the second bearing **1105** can facilitate movement of the forming roll **901**, for example, when a first end **909** is moved independently of the second end **911**, or when the second end **911** is moved independently of the first end **909**.

[0090] In some embodiments, the glass manufacturing apparatus **100** can provide several benefits associated with manufacturing the glass ribbon **123**. For example, one or more of the first translational drive apparatus **241** or the second translational drive apparatus **243** can comprise a servo motor, while the other of the first translational drive apparatus **241** or the second translational drive apparatus **243** can comprise a pneumatic cylinder or a servo motor. With the first translational drive apparatus **241** and/or the second translational drive apparatus **243** comprising the servo motor, more incremental control of the movement and position of the first forming roll **117** and the second forming roll **119** can be achieved, thus facilitating a more accurate gap width between the first forming roll **117** and the second forming roll **119**. Similarly, the servo motor can facilitate positional adjustment of the first forming roll **117** and/or the second forming roll **119** during operation of the glass manufacturing apparatus **100**. For example, the servo motors can adjust the position of the first forming roll **117** relative to the second forming roll **119** and/or the position of the second forming roll **119** relative to the first forming roll **117** while the glass manufacturing apparatus **100** is in operation and the glass ribbon **105** is being delivered to the first forming roll **117** and the second forming roll **119**, thus reducing downtime and increasing efficiency. In some embodiments, when the second translational drive apparatus **243** comprises the pneumatic cylinder, the second forming roll **119** can allow for a solidified piece of material to pass through the gap **121** between the first forming roll **117** and the second forming roll **119** by moving away from the first forming roll **117**, which can reduce the likelihood of damage to the first forming roll **117** and the second forming roll **119**.

[0091] In addition, or in the alternative, the drive apparatus **129** can facilitate a more accurate gap width between the first forming roll **117** and the second forming roll **119**. For example, when the first translational drive apparatus **241** comprises the first end translational drive apparatus **245** and the second end translational drive apparatus **247**, the first end **205** and the second end **207** of the first forming roll **117** can be moved independently of one another. This independent movement can facilitate adjustment of the gap width along the length of the first forming roll **117**, which may be beneficial when the first forming roll **117** and/or the second forming roll **119** comprise variations in cross-sectional size. Similarly, when the second translational drive apparatus **243** comprises the third end translational drive apparatus **265** and the fourth end translational drive apparatus **267**, the first end **225** and the second end **227** of the second forming roll **119** can be moved independently of one another. This independent movement can facilitate adjustment of the gap width along the length of the second forming roll **119**, which may be beneficial when the first forming roll **117** and/or the second forming roll **119** comprise variations in cross-sectional size. To reduce variations in cross-sectional size of the forming rolls **117**, **119**, one or more surfaces of the forming rolls **117**, **119** can be machined following assembly of the forming rolls **117**, **119**, wherein the machining can reduce a variation in the width of the gap **121** defined between the first forming roll **117** and the second forming roll **119**.

[0092] As used herein the terms "the," "a," or "an," mean "one or more," and should not be limited to "only one" unless explicitly indicated to the contrary. Thus, for example, reference to "a component" includes embodiments having two or more such components unless the context clearly indicates otherwise.

[0093] As used herein, the term "about" means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term "about" is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites "about," the numerical value or end-point of a range is intended to include two embodiments: one modified by "about," and one not modified by "about." It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0094] The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, as defined above, “substantially similar” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially similar” may denote values within about 10% of each other, for example within about 5% of each other, or within about 2% of each other.
[0095] As used herein, the terms “comprising” and “including,” and variations thereof shall be construed as synonymous and open-ended, unless otherwise indicated.
[0096] While various embodiments have been described in detail relative to certain illustrative and specific embodiments thereof, the present disclosure should not be considered limited to such, as numerous modifications and combinations of the disclosed features are envisioned without departing from the scope of the following claims.

Claims

1. A method of manufacturing a glass ribbon comprising: introducing the glass ribbon along a travel path in a travel direction to a gap defined between a first forming roll and a second forming roll; passing the glass ribbon through the gap; and changing a width of the gap by moving one or more of the first forming roll independently of the second forming roll along a movement axis that is substantially perpendicular to the travel path or the second forming roll independently of the first forming roll along the movement axis.
2. The method of claim 1, further comprising assembling the first forming roll and the second forming roll, and, after the assembling, machining one or more surfaces of the first forming roll or the second forming roll to reduce a variation in the width of the gap.
3. The method of claim 1, wherein the changing the width of the gap comprises moving one end of the first forming roll to accommodate for a variation in the width of the gap along the length of the gap.
4. The method of claim 1, wherein the changing the width of the gap occurs as the glass ribbon is received within the gap.
5. The method of claim 1, further comprising monitoring a characteristic of the glass ribbon and, based on the characteristic, changing the width of the gap.
6. The method of claim 5, wherein the characteristic comprises one or more of a force exerted on one or more of the first forming roll or the second forming roll, or a thickness of the glass ribbon.
7. A method of manufacturing a glass ribbon comprising: assembling a first forming roll and a second forming roll; machining one or more surfaces of the first forming roll or the second forming roll to reduce a variation in a width of a gap defined between the first forming roll and the second forming roll; introducing the glass ribbon along a travel path in a travel direction to the gap; and passing the glass ribbon through the gap.
8. The method of claim 7, wherein the assembling comprises attaching a first shaft to a first side of a first roller and a second shaft to a second side of the first roller to form the first forming roll, and attaching a third shaft to a first side of a second roller and a fourth shaft to a second side of the second roller to form the second forming roll.
9. The method of claim 8, wherein the assembling comprises attaching the first shaft to a first bearing and the second shaft to a second bearing and attaching the third shaft to a third bearing and the fourth shaft to a fourth bearing.
10. The method of claim 7, wherein the machining occurs after the assembling of the first forming roll and the second forming roll.